Megaloblastic Anemia

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Introduction

- ☑ Megaloblastic anemia is caused by a deficiency of vitamin B12 and folate in 95% of cases.
- ☑ Rarely, it can be caused by inborn errors of metabolism of vitamin B12 and folate **(Table 1)**.
- ☑ It is named because of the classical presence of megaloblasts in bone marrow and macrocytes (large, structurally abnormal, and immature red blood cells) in the blood. These are formed as a result of asynchronous maturation between the nucleus and cytoplasm due to defective deoxyribonucleic acid (DNA) synthesis.

TABLE 1

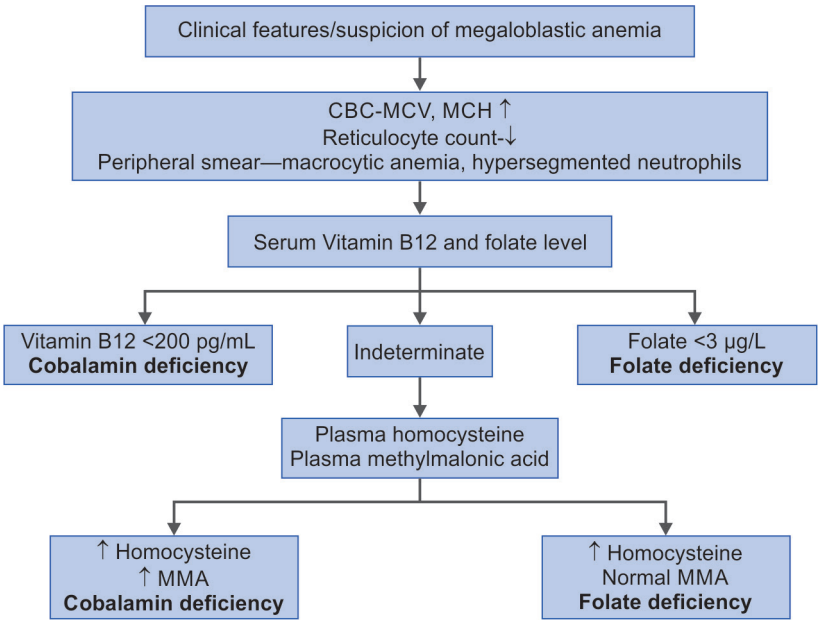
Causes of deficiency	***Vitamin B12***	***Folate***
Inadequate intake	Dietary (<2 mg/day)—lactovegetarians, lacto-ovo-vegetarians, vegans, maternal deficiency	Maternal deficiencies with prolonged breastfeeding, goat's milk feeding, prematurity Increased requirement—chronic hemolysis
Defective absorption	Congenital intrinsic factor deficiency, pernicious anemia, Imerslund–Gräsbeck syndrome, malabsorption due to various intestinal pathologies	Congenital Acquired—postintestinal surgeries, antibiotics, and other drugs
Defective transport	Transcobalamin deficiency	*FOLR1* gene mutation
Disorders of metabolism	Methylcobalamin deficiency, liver disease, protein energy malnutrition	Congenital—methylenetetrahydrofolate reductase deficiency Acquired—vitamin B12 deficiency and liver disease

Clinical Features

- ☑ Lethargy, easy fatigability, and anorexia
- ☑ Pallor and sallow complexion
- ☑ Glossitis
- ☑ Hyperpigmented nail beds and knuckles
- ☑ Neurological manifestations—infantile tremor syndrome, hypotonia, paresthesias, seizures, and delayed development
- ☑ Petechiae and other hemorrhagic manifestations
- ☑ Hepatosplenomegaly (30–40% of cases) mimicking acute leukemia
- ☑ Thrombosis may occur in cases with hyperhomocysteinemia.

Workup

Flowchart 1 presents basic laboratory investigations.



(CBC: complete blood count; MCH: mean corpuscular hemoglobin; MCV: mean corpuscular volume)

Points to Note

- ☑ *Elevated lactate dehydrogenase (LDH)* is a marker of ineffective erythropoiesis and often seen in megaloblastic anemia.
- ☑ A *mean corpuscular volume (MCV) > 110* is highly suggestive of megaloblastic anemia.
- ☑ Hypersegmented neutrophils—*more than five lobes* in at least 5% of neutrophils or a single cell showing six or more lobes is a cardinal feature.
- ☑ Patients may present with *pancytopenia* with features indistinguishable from aplastic anemia—bone marrow examination will help in these cases.

Workup

Special Investigations

- ☑ Bone marrow examination findings—hypercellularity with increase in all three marrow elements, especially erythroid compartment. Megaloblastic erythropoiesis, abnormal leucopoiesis, and megakaryocytes may be normal or increased.
- ☑ To confirm pernicious anemia—intrinsic factor blocking antibody test.
- ☑ When common causes are excluded, urine for orotic acid and tandem mass spectrometry for metabolic disorders should be done.
- ☑ Holotranscobalamin (HoloTC) is the "active" fraction of serum cobalamin and is more specific than serum cobalamin levels. It may be used as a routine diagnostic laboratory in the future.

Management

- ☑ Vitamin B12 and folic acid can be given either orally or parenterally.
- ☑ Parenteral administration is necessary in patients with malabsorption due to various causes, or intrinsic factor deficiency.
- ☑ Intravenous route should be used in patients with thrombocytopenia.
- ☑ Lifelong supplementation is required in patients with inborn errors of metabolism.
- ☑ In cases of confirmed deficiency, cobalamin deficiency should be corrected first for a few days before starting folate, as folate alone may precipitate or aggravate neurologic symptoms, if present.

Cobalamin

The dosages, schedules, and formulations used (cyanocobalamin, methylcobalamin, and hydroxycobalamin) widely vary in individual practice, but broad recommendations are as follows:

- ☑ 500–1,000 µg/day orally daily for 4–6 weeks, then once a week (or)
- ☑ 1,000 µg/day intramuscular/intravenous for 2 weeks or thrice weekly for 2 weeks followed by weekly for another six doses till hematocrit returns to normal.

Folic Acid

- ☑ <6 months of age—50 µg/day
- ☑ 7 months to 13 years—1 mg/day for 2–3 weeks, then 0.1–0.5 mg/day
- ☑ >13 years—1 mg/day for 2–3 weeks, then 0.5 mg/day

Response to Treatment

- ☑ Symptomatic response is usually seen within 48–72 hours.
- ☑ Brisk reticulocytosis within 2–3 days, peaking at 5–8 days—degree of response is inversely proportional to degree of anemia.
- ☑ Serum LDH, homocysteine, and methylmalonic acid (MMA) normalize within a week.
- ☑ Bone marrow ineffective erythropoiesis is reversed within 72 hours.
- ☑ Hemoglobin levels normalize within 4–8 weeks.

Thiamine-responsive Megaloblastic Anemia

- ☑ Thiamine-responsive megaloblastic anemia syndrome (TRMA) is characterized by megaloblastic anemia, progressive sensorineural hearing loss, and diabetes mellitus.
- ☑ It is caused by a mutation in the *SLC19A2* gene and inherited in autosomal recessive fashion.
- ☑ Onset of megaloblastic anemia occurs between infancy and adolescence. The anemia is corrected with high doses of thiamine (vitamin B1), but the red cells remain macrocytic, and anemia can recur when treatment is withdrawn.
- ☑ Lifelong supplementation of thiamine (50–100 mg/day) is required.

Further Readings

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Under 5 Wheeze

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